Answer Key & Teacher Notes

Session 1 — Plotting Points on the Coordinate Plane · Session 2 — Reflecting Points Across Axes

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Plotting Points on the Coordinate Plane

TALK PROMPTS

- How do the signs inside an ordered pair tell you which quadrant a point lands in?
- Why does a point with a zero coordinate land on an axis instead of inside a quadrant?

PARTNER TASK

City Map. One partner plots the museum at (−4, −7) and the stadium at (9, 1); the other names each quadrant and describes the moves from the origin. Switch roles with the Park Map: the Pond at (−8, 4) and the Swing Set at (6, −5).

WATCH FOR

- Moving along the y-axis first (plotting (−5, 4) as down 5, right 4).
- Switching x and y, thinking (−2, 6) and (6, −2) are the same point.
- Naming a quadrant for a point on an axis, such as (0, −5).

SESSION 2 · TEACHER NOTES

Reflecting Points Across Axes

TALK PROMPTS

- Why does only one coordinate change sign when a point reflects across an axis?
- How can you name the image before you plot a single point on the grid?

PARTNER TASK

Two Kites. One partner plots Kite A at (−6, 3) and names its reflection across the y-axis; the other finds the left-right distance between the kites (12 units). Switch roles with the Stage Lights: (−8, 12) across the x-axis.

WATCH FOR

- Changing both signs for a single reflection ((−1, 8) written as (1, −8)).
- Changing the wrong coordinate (across the y-axis, changing y).
- Swapping the two coordinates instead of changing a sign.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.C.6c	1	2	1	2	2	2	2	3	2	1	1
S1 · Form B	6.NS.C.6c	1	2	1	2	2	2	2	3	2	1	1
S2 · Form A	6.NS.C.6b	1	2	1	1	2	2	2	3	2	1	1
S2 · Form B	6.NS.C.6b	1	2	1	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

coordinate plane: A grid made by a horizontal and a vertical **number line**.

ordered pair: Two numbers (x, y) that name one **point**.

origin: The point **(0, 0)** where the two axes cross.

quadrant: One of the **four** sections of the plane.

- The x-coordinate tells you to move **right or left**. The y-coordinate tells you to move up or down.
- Signs name the quadrant: (−, +) is Quadrant **II** and (−, −) is Quadrant III.

Example: Start at the origin, (0, 0). Move 5 units **left**, because x = −5. Move 4 units **down**, because y = −4. Answer: (−, −) means Quadrant **III**, the lower left.

YOU TRY

1. a. Quadrant IV b. Quadrant II

2. A: 4 left, 2 up (Quadrant II); B: 3 right, 5 down (Quadrant IV)

EXIT TICKET

1. a. Quadrant III b. Quadrant I

2. (−5, 2)

3. **B** ((3, −4))

PRACTICE FORM A

1. a. II b. I c. III d. IV e. on the y-axis f. on the x-axis

2. a. Gate 6 right, 3 up; Pond 4 left, 8 up; Bench 7 left, 2 down; Swing Set 5 right, 6 down b. Gate I, Pond II, Bench III, Swing Set IV c. (0, 0)

3. (−6, 11), Quadrant II

4. (−5, 1) is in II; (1, −5) is in IV; not the same point

5. No. (−2, 6) is 2 left and 6 up (Quadrant II). (6, −2) is 6 right and 2 down (Quadrant IV). Order matters.

6. (−3.5, 2.5) in Quadrant II; (4.5, −1.5) in Quadrant IV

7. a. museum III, stadium I b. (0, 0)

8. a. school I, park II, pool III b. 3 (school and park) c. 6 units

9. The first number is x, so −5 means 5 units LEFT. The second number is y, so 4 means 4 units UP. The point (−5, 4) is in Quadrant II.

10. **B** (Quadrant II)

11. **B** (right or left)

Self-check word **ORIGIN**

PRACTICE FORM B

1. a. II b. I c. III d. IV e. on the y-axis f. on the x-axis

2. a. Gate 3 right, 6 up; Pond 8 left, 4 up; Bench 2 left, 7 down; Swing Set 6 right, 5 down b. Gate I, Pond II, Bench III, Swing Set IV c. (0, 0)

3. (−9, −5), Quadrant III

4. (−4, 2) is in II; (2, −4) is in IV; not the same point

5. No. (3, −7) is 3 right and 7 down (Quadrant IV). (−7, 3) is 7 left and 3 up (Quadrant II). Order matters.

6. (−2.5, −4.5) in Quadrant III; (3.5, 1.5) in Quadrant I

7. a. museum IV, stadium II b. (0, 0)

8. a. school IV, park III, pool II b. −2 (school and park) c. 6 units

9. The first number is x, so −3 means 3 units LEFT. The second number is y, so 6 means 6 units UP. The point (−3, 6) is in Quadrant II.

10. **D** (Quadrant IV)

11. **C** (Quadrant III)

Self-check word **POINTS**

SESSION 2 · ANSWERS

Reflecting Points Across Axes

GUIDED NOTES

reflection: A **flip** that makes a mirror image across a line.

x-axis: The **horizontal** number line through the origin.

y-axis: The **vertical** number line through the origin.

- Across the x-axis, the **y**-coordinate changes sign.
- Across the y-axis, the **x**-coordinate changes sign.

Example: Across the x-axis: keep x = 7 and change −2 to **2**. Image: (7, 2). Across the y-axis: keep y = −2 and change 7 to **−7**. Answer: The images are (7, 2) and **(−7, −2)**.

YOU TRY

1. a. (3, 5) b. (6, 1)

2. B = (−2, −4)

EXIT TICKET

1. a. (4, −9) b. (8, −3)

2. the x-axis; 12 units apart

3. **B** ((−3, −7))

PRACTICE FORM A

1. a. (5, −3) b. (2, 8) c. (7, 1) d. (9, −4)

2. a. (4, −7), Quadrant IV b. (−4, 7), Quadrant II

3. (6, 3); 12 units

4. (−8, −12); 24 units

5. the x-axis; x stays 2 and only y changes sign

6. He changed y instead of x. Correct image: (3, 6)

7. a. (5, 8) b. (−5, 8) c. Quadrant II

8. a. (−3, 4) b. (3, −4) c. (−3, −4): reflect A across both axes (change both signs)

9. A reflection across the x-axis changes only the y-coordinate. x stays −1 and 8 becomes −8. Correct image: (−1, −8).

10. **B** ((4, 6))

11. **A** (the x-coordinate)

Self-check word **REFLECTS**

PRACTICE FORM B

1. a. (6, −2) b. (5, 7) c. (8, 3) d. (10, −5)

2. a. (−5, −6), Quadrant III b. (5, 6), Quadrant I

3. (7, 2); 14 units

4. (−6, −9); 18 units

5. the y-axis; y stays 7 and only x changes sign

6. She changed y instead of x. Correct image: (−4, −5)

7. a. (−6, 3) b. (6, 3) c. Quadrant I

8. a. (−5, 2) b. (5, −2) c. (−5, −2): reflect A across both axes (change both signs)

9. A reflection across the x-axis changes only the y-coordinate. x stays −7 and 2 becomes −2. Correct image: (−7, −2).

10. **B** ((5, −3))

11. **A** ((2, 8))

Self-check word **SYMMETRY**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · PLOTTING POINTS ON THE COORDINATE PLANE

1. A (4, 3) I; B (−4, 2) II; C (−2, −4) III; D (3, −4) IV

2. P(2, 4) in I; Q(−4, −1) in III; S(−3.5, 3) in II; G(4, −2.5) in IV. a. the pool b. the garden

3. a. Quadrant III b. Quadrant IV c. on the y-axis d. Quadrant II e. on the x-axis f. Quadrant IV

4. No. Order matters. (−3, 5) is 3 left and 5 up (Quadrant II). (5, −3) is 5 right and 3 down (Quadrant IV).

SHEET 2 · REFLECTING POINTS ACROSS AXES

1. a. B = (−4, −3) b. C = (4, 3)

2. a. (6, −2) b. (4, −7)

3. (7, 2.5)

4. K = (−6, 5). Reflecting across the x-axis flips the sign of y, so flip it back: −5 becomes 5.

5. He changed the y-coordinate. Across the y-axis, y stays the same and x changes sign. Correct point: (3, −8).

6. **B** ((5, 2))

7. **A** (the x-axis)

October 2026 conceptual additions — teacher key

Session 1 · Plotting Points on the Coordinate Plane

(−2,3) is Quadrant II; (0,−3) is on the y-axis; origin on both axes. Quadrants are the open regions between axes.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: (−2,3) is Quadrant II; (0,−3) is on the y-axis; origin on both axes. Quadrants are the open regions between axes.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Reflecting Points Across Axes

Separate images: (3,2) and (−3,−2). Sequential final image: (−3,2). Point (0,4) stays fixed under reflection across the y-axis.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Separate images: (3,2) and (−3,−2). Sequential final image: (−3,2). Point (0,4) stays fixed under reflection across the y-axis.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.